



Task 1

- Create an electrical circuit of the power grid, describing the process of electricity transmission from generation to the consumer. Describe the components shown in the circuit and their functions. Include substation components in the correct order and explain the operating principles of the power grid.
- What is *Baltic RCC*'s role in ensuring the reliability and security of the power grid in the Baltic states?

Summarize all your findings and results in a presentation (no longer than 10-15 minutes).

Task 2

Request data from the website of the *Elering* regulation market here: <https://baltic.transparency-dashboard.eu/> and draw a graph based on returned data and give an assessment of regulation activities. Target day is 22.09.2025.

1. Request activated aFRR volumes (afrr_activation) and total imbalance volumes (imbalance_volumes_v2). API documentation available here: <https://baltic.transparency-dashboard.eu/documentation/api>
2. Make a graph and additional metrics to support the assessment of the amount of activation of aFRR reserves in relation to the imbalance. Provide also reasoning and background for the assessment.

Hints:

- The *Python* modules requests, pandas and matplotlib are helpful.
- API documentation is here: <https://baltic.transparency-dashboard.eu/documentation/api>
- Theory on aFRR here: <https://www.next-kraftwerke.com/knowledge/afrr>

Result:

1. Code that retrieves the data from dashboard and makes a graph from the data. Shared either as a text file or a link to *GitHub* repository (preferred).
2. Reasoning and background for the assessment with theoretical background using different metrics.



Task 3

Assess the provided EQ profile instance (CGMES model format in *xml* containing information on network physical elements).

1. What is the total production capacity of the generators in the model?
2. What are the nominal voltages of the windings of the transformer NL_TR2_2 (ID: _2184f365-8cd5-4b5d-8a28-9d68603bb6a4)?
3. What is the permanently allowed limit for line segment NL-Line_5 (ID: _e8acf6b6-99cb-45ad-b8dc-16c7866a4ddc) and temporarily allowed? What is the difference between those limits?
4. Which generator is set as slack in the model? Why does the model need a slack node?
5. Find mistakes in the model (semantic, power system related, and logical errors are all present).

Hints:

- It is suggested to use *Notepad++* for finding elements or *Python* (*Python* is preferred). If you use *Python*, please provide sample code with the task.
- The *Python* modules *triplets* or *rdflib* can be helpful.
- CGMES structure and principles to help if stuck are explained also here: <https://www.youtube.com/watch?v=FjpyO0iWhWo>

Result:

1. Answers to the questions with image snips to support the answers.
2. If *Python* code is used (preferred), provide the code via a text file or, preferably, a link to a *GitHub* repository.